RFP vs Existing Project — Gap Analysis

BUIDCO Integrated Project Monitoring Dashboard · prepared 3 October 2026

1. Scope of this analysis

RFP source: Bihar Urban Infrastructure Development Corporation Ltd., *Request for Proposal* — pages 22–23, **Clause 5: "Development of Integrated Project Monitoring Dashboard"**, together with the adjacent consultant obligations in Clause 4(c)–(e) (Financial & Payment Management, Claims/Variations/Disputes, Risk Management & Reporting).

System assessed: the existing BUIDCO Project Monitoring Dashboard — React + Vite frontend, Node.js/Express + TypeScript API, PostgreSQL via Drizzle ORM. 23 database tables, 11 API route groups, 14 services, 21 UI pages, 4 user roles.

This exercise was performed strictly **read-only**. No source file, database object, API, configuration or behaviour of the existing application was modified. Every status below is evidenced against actual code or schema, not assumed.

✅ Completed	🟡 Partial	❌ Missing	🔄 Modify
Implemented and working	Exists but short of the RFP wording	No implementation or data model	Exists, needs rework to comply

2. Headline result

11	11	10	32	~34%
COMPLETED	PARTIAL	MISSING	REQUIREMENTS	FULLY MET

The dashboard is strong exactly where the RFP asks for **project, progress and financial monitoring** — those modules are largely built and in use. The gaps cluster in three places:

- **Geospatial** — GIS visualisation is required but there is no map, and no coordinate data to put on one.
- **Contract risk instruments** — insurance validity, claims, disputes and a risk register are not modelled at all.
- **Operational governance** — backup/disaster recovery, escalation routing, push alerting, manuals and support are absent.

> **Compliance risk to raise early.** Clause 5(b)(vii) requires a "secure hosting environment". The API is currently on a free hosting tier that suspends the service after ~15 minutes of inactivity; the next request takes 33–56 seconds and the frontend proxy times out before it answers. This is reproducible and was measured repeatedly. As it stands the deployment would not satisfy an availability review, independent of any feature work.

3. Clause 5(a) — System Design and Architecture

RFP requirement	Status	Evidence / gap
Requirement assessment in consultation with BUIDCO	❌	Process deliverable. No requirement-assessment record exists in the repository.
Secure, web-based, role-based access-enabled dashboard	✅	React SPA over an Express REST API. RBAC with 4 roles (MD/Admin/PD/Viewer), per-user create/update/delete flags, PD scoped to division. Enforced server-side in middleware.
Compatibility and integration with existing MIS/ERP systems	❌	No integration layer, connector, or external API client. Only inbound Excel import exists.
Data validation and governance protocols	🟡	Validation is solid — Zod schemas at the API edge, reused by the bulk importer. Governance protocols (data dictionary, ownership, retention, correction policy) are not documented.

4. Clause 5(b)(i) — Project Overview Module

RFP requirement	Status	Evidence / gap
Sector-wise and project-wise details	✅	Overview, Projects, Sectors, Divisions and Districts pages; `sector` / `scheme` master tables with per-sector summary cards.
Approved cost, awarded cost, funding source, contractor details	✅	`aa_amount_cr` (approved), `agreement_amount_cr` (awarded), `funding_source` enum, `contractor`. All on the project record and in the table.
Start date, completion date, and current status	✅	`sanction_date`, `appointed_date`, `planned_end_date`, `revised_end_date`, `expected_completion_date`, `status`.

5. Clause 5(b)(ii) — Physical Progress Monitoring

RFP requirement	Status	Evidence / gap
Milestone tracking	✅	`project_milestone` + `milestone_progress` tables, Milestones section in the Input Sheet, milestone-history endpoint.
Planned vs actual progress comparison	✅	`physical_progress_pct` vs `scheduled_progress_pct`; physical-vs-financial chart on the Overview dashboard.
Delay analysis and critical alerts	✅	Delay-days computation, delay categories, "Delayed Projects" window filter (30/60/90 days, 6 months, 1 year), delay reason and department-stuck-at fields.
GIS-enabled visualization (where applicable)	❌	**Not implemented, and no data to implement it from.** No mapping library is installed. The `geo_photo` table stores url, caption, date, source and filename — **no latitude or longitude columns**. The "Geo Photos" route is a placeholder marked *deferred*.

6. Clause 5(b)(iii) — Financial Monitoring

RFP requirement	Status	Evidence / gap
Budget allocation vs expenditure	✅	Funds & UC module: `opening_balance_cr`, `grant_received_cr`, `expenditure_incurred_cr`, central/state share split.
Fund utilization tracking	✅	Funds & UC page, `uc_submitted_date`, `sanction_no`, and a portfolio financial-utilisation KPI.
Cost overrun indicators	✅	Computed Cost Overrun and Time Overrun fields in the project profile (AA vs revised AA vs agreement value).
Payment status tracking	🟡	Fields exist — `total_payments_made_cr`, `last_payment_date`, `last_ra_bill_no`, `retention_money_held_cr`. What is missing is a **payment lifecycle**: no IPC record, no bill submitted → verified → certified → paid states, no ageing of pending bills. Clause 4(c) expects IPC scrutiny and final bills.

7. Clause 5(b)(iv) — Contract Management Module

RFP requirement	Status	Evidence / gap
Contract duration tracking	✅	`agreement_date`, `appointed_date`, planned/revised end dates, computed time overrun.
Performance security and insurance validity alerts	🟡	Performance security is **done well** — `pbg_number`, `pbg_amount_cr`, `pbg_expiry_date`, a "PBG Expiring ≤ 30 Days" dashboard card, a colour-graded column, and a database view. **Insurance is entirely absent** — no field, no alert, zero references anywhere in the codebase.
Variation orders and claims tracking	🟡	Variations are covered by the CoS/EoT module (`cos_eot_item` with `category` incl. QUANTITY VARIATION, `cos_amount_cr`, `variation_pct`). **Contractor claims are not modelled** — Clause 4(d) requires a Claims Register.
EoT and dispute status tracking	🟡	EoT is complete — `eot_number`, `eot_days_granted`, `original_end_date`, `new_end_date`, time-linked flag. **Dispute status is not modelled**; "dispute" appears only inside free-text remarks. Clause 4(d) requires a Disputes Register and arbitration support.

8. Clause 5(b)(v) — Risk and Alert System

RFP requirement	Status	Evidence / gap
Automated alerts — delays, cost overruns, expiring securities, critical milestones	🟡	All four conditions *are* detected and shown: delay windows, cost-overrun indicator, PBG and O&M expiry cards, milestone tracking. But they are **passive, on-screen** indicators. There is no push mechanism — no email or SMS alert, no scheduled job, no notification record. "Automated" in the RFP sense is not met.
Red-Amber-Green (RAG) status indicators	🟡	Colour coding exists and is used consistently (red/orange/yellow/green badges for status, priority, PBG and O&M). But there is no formal three-state RAG model, no defined thresholds per metric, and no RAG legend or roll-up at project or portfolio level.
Escalation matrix functionality	❌	Not implemented. No escalation rules, no severity tiers, no owner hierarchy, no time-based routing. The only code matches are unrelated word usage in remark-tone tests.

9. Clause 5(b)(vi) — Reporting and Analytics

RFP requirement	Status	Evidence / gap
Customizable reports (project-wise, sector-wise, district-wise)	🟡	All three cuts exist as pages (Projects, Sectors, District-wise Summary, plus Divisions), with column show/hide, per-column filters and sorting. What is missing is **"customizable" in the report-builder sense**: no user-defined report definitions, no saved/named reports, no scheduled generation.
Graphical dashboards and trend analysis	🟡	Graphical dashboards: yes — KPI tiles, physical-vs-financial bar chart by scheme, status donut, stage buckets. **Trend analysis is thin** — a physical-progress history endpoint exists but there is no time-series trend view, no S-curve, no period-on-period comparison.
Export functionality (PDF/Excel formats)	✅	Exceeds the requirement. Excel via ExcelJS, PDF via jsPDF + autotable, and PowerPoint via PptxGenJS, across project, table and MD-summary exporters.

10. Clause 5(b)(vii) — Data Security and Governance

RFP requirement	Status	Evidence / gap
Role-based access control	✅	Four roles plus granular per-user permissions; PD restricted to assigned divisions; cross-division requests return 404 rather than leaking existence. Enforced in backend middleware, not only hidden in the UI.
Secure hosting environment	🟡	HTTPS, Helmet security headers, CORS allowlist, bcrypt password hashing, JWT with rotating refresh tokens, Redis-backed rate limiting. **But** the API runs on a free tier that sleeps after ~15 min, giving 33–56s cold starts and user-visible failures. No security review, hardening checklist or penetration test on record.
Data backup and disaster recovery mechanisms	❌	Nothing in the repository: no backup schedule, no retention policy, no restore procedure, no RPO/RTO statement, no tested recovery. Zero references to backup or disaster recovery.
Audit trails for all data modifications	✅	`audit_log` (who / what / when) plus `audit_log_change` with old and new values per field, surfaced in an Audit Trail page. Genuinely strong against this clause.

11. Clause 5(b)(viii) — Capacity Building and Support

RFP requirement	Status	Evidence / gap
Training sessions for BUIDCO officials and staff	❌	Process deliverable; no training plan, material or attendance record exists.
User Manuals and Standard Operating Procedures (SOPs)	🟡	An in-app KPI guide drawer exists and is genuinely useful, and project presentation notes were produced. Neither is a role-wise User Manual or an SOP set.
Technical support during the agreed maintenance period	❌	No support model, SLA, ticketing route or escalation contact defined.

12. Clause 4 obligations that depend on dashboard data

RFP requirement	Status	Evidence / gap
4(c) IPC scrutiny and final bill certification	❌	No IPC entity or certification workflow. Payment data is summary-level only.
4(d) Claims and Disputes Register	❌	Neither register exists as a data structure.
4(e) Risk Register	❌	No risk entity, no likelihood/impact scoring, no mitigation owner or review date.
4(e) Periodic Contract Management Reports (monthly/quarterly)	🟡	Minutes of Meeting with action points, management actions and MPR month fields exist, but there is no scheduled monthly/quarterly report generation or distribution.

> **Worth saying plainly in any submission:** the areas the RFP weights most heavily for day-to-day monitoring — project register, physical progress, financial tracking, exports, RBAC and audit — are built to a good standard. The gaps are real but mostly additive; only GIS and the payment lifecycle require new data modelling of any size.

Pending Work — Prioritised

Every item traces to an RFP clause. Nothing below has been implemented.

13. Critical / Mandatory

Explicitly required by Clause 5 and currently absent. These are compliance failures, not enhancements.

#	Task	RFP ref	Why critical
C1	**Hosting and availability remediation** — move the API to an always-on instance; document uptime and an availability target.	5(b)(vii)	Service currently sleeps and fails user requests. Fails "secure hosting environment" on availability grounds. Cheapest fix on this list.
C2	**Backup and disaster recovery** — automated database backups, defined retention, documented and *tested* restore, stated RPO/RTO.	5(b)(vii)	Explicitly named in the RFP and entirely absent. Also a data-loss risk today.
C3	**GIS-enabled visualization** — add latitude/longitude to projects and geo-photos, then a map view with sector/status filters.	5(b)(ii)	Named module requirement with no implementation and no underlying data. Largest single build.
C4	**Insurance validity tracking and alerts** — policy number, insurer, cover, validity dates, expiry alerts alongside PBG.	5(b)(iv)	Named in the same breath as performance security; PBG is done, insurance has no field at all.
C5	**Automated (push) alerting** — scheduled job that emails/notifies owners on delays, cost overruns, expiring securities and critical milestones.	5(b)(v)	Detection already exists; only the delivery mechanism is missing. Converts passive screens into the "automated alerts" the RFP asks for.
C6	**Escalation matrix** — severity tiers, owner hierarchy, time-based escalation, escalation log.	5(b)(v)	Named functionality, zero implementation.
C7	**Claims and Disputes Register** — claim/dispute records with status, value, dates, linkage to project and CoS/EoT.	5(b)(iv), 4(d)	Required by two clauses; currently only free text in remarks.
C8	**Risk Register** — risk entity with likelihood, impact, mitigation, owner, review date, feeding the dashboard.	4(e)	Explicit consultant deliverable with no system support.

14. High Priority

#	Task	RFP ref	Why
H1	**Payment lifecycle / IPC tracking** — IPC records with submitted → verified → certified → paid states, ageing of pending bills.	5(b)(iii), 4(c)	Turns summary payment fields into the certification workflow the RFP describes. Directly addresses the complaint that sank the earlier state system.
H2	**Formal RAG framework** — defined thresholds per metric, consistent three-state model, project and portfolio roll-up, visible legend.	5(b)(v)	Colour coding exists but is not a defensible RAG scheme.
H3	**Report builder** — user-defined, saveable, named report definitions over the existing filters; scheduled generation.	5(b)(vi)	"Customizable reports" currently means ad-hoc filtering only.
H4	**Trend analysis** — time-series of physical and financial progress, S-curve vs plan, period-on-period comparison.	5(b)(vi)	History data is captured but never visualised as a trend.
H5	**User Manual and SOPs** — role-wise manual (MD / Admin / PD / Viewer) and SOPs for data entry, approval and month-end.	5(b)(viii)	Named deliverable. Also the exact failure mode that derailed the state's earlier works system.
H6	**Monthly/quarterly Contract Management Report** — scheduled, exportable, standard format.	4(e)	Named periodic deliverable.

Pending Work (continued)

Medium and low priority items, and the recommended build order

15. Medium Priority

#	Task	RFP ref	Why
M1	**MIS/ERP integration capability** — documented REST interface and/or connector, with authentication for machine clients.	5(a)	Qualified "if any" in the RFP, so scope depends on what BUIDCO actually runs — confirm before building.
M2	**Data governance protocol** — data dictionary, field ownership, correction policy, retention schedule.	5(a)	Validation is strong; governance is undocumented.
M3	**Training programme** — plan, materials, sessions, attendance record.	5(b)(viii)	Process deliverable; schedule near go-live.
M4	**Support model and SLA** — channel, response/resolution targets, maintenance-period cover.	5(b)(viii)	Contractual obligation to define.
M5	**Security review** — hardening checklist, dependency audit, penetration test.	5(b)(vii)	Strengthens the "secure" claim with evidence.

16. Low Priority / Enhancement

#	Task	RFP ref	Why
L1	Documented requirement-assessment record, signed off with BUIDCO.	5(a)	Retrospective paperwork; closes an audit point cheaply.
L2	Bundle-size and performance optimisation (main JS bundle is ~1.2 MB).	—	Not an RFP requirement; improves usability on slow government connections.

17. Recommended implementation sequence

Ordered by dependency and by what de-risks the engagement earliest.

Stage	Items	Rationale
Stage 1 Stabilise	C1 hosting/availability · C2 backup & DR · M5 security review	Non-negotiable platform hygiene, independent of features. C1 is quick and removes a visible, demonstrable failure. Do this before any demonstration to the client.
Stage 2 Close named module gaps	C4 insurance · C7 claims & disputes · C8 risk register	Pure additive data modelling next to tables that already exist (PBG, CoS/EoT). Low risk to current behaviour, and each closes a named clause.
Stage 3 Make alerting real	C5 push alerts · C6 escalation matrix · H2 RAG framework	Depends on Stage 2 so that insurance, claims and risks can raise alerts too. Detection logic already exists, so this is mostly delivery and rules.
Stage 4 Geospatial	C3 GIS visualisation	Largest build and needs new coordinate capture plus back-filling existing projects. Start the data collection early even if the map is built late.
Stage 5 Financial depth	H1 payment/IPC lifecycle · H6 periodic reports	The heaviest workflow change. Sequence after alerting so that pending-bill ageing can feed the alert engine.
Stage 6 Analytics	H3 report builder · H4 trend analysis	Best built once all entities exist, so reports can span them.
Stage 7 Handover	H5 manuals & SOPs · M3 training · M4 support/SLA · M2 governance · L1 requirements record	Documentation and enablement must reflect the finished system, so they land last — but draft alongside, do not leave to the final week.

Risk Precedent & Compliance Summary

What went wrong last time, and where the project stands clause by clause

18. A warning worth heeding

Bihar's earlier state-wide works system (WAMIS) was launched and then suspended within a month. The reported causes were not missing features:

- **Contractor bills were not paid on time** through the new system — which is precisely what task **H1 (payment/IPC lifecycle)** addresses. Treat it as higher risk than its "High" label suggests.
- **Training was inadequate** — nominated departmental representatives attended but did not learn the system, and so could not train their own staff. That is tasks **H5** and **M3**, and it argues for naming nodal officers who genuinely know departmental and field-level procedure.
- There was a stated risk of **violating the Bihar Public Works Code** — worth an explicit compliance check of workflows against the Code before go-live.

Source: Dainik Jagran, 27 August 2022. Supplied alongside the RFP; it is background, not a requirement document, but the failure modes map directly onto tasks already on this list.

19. Summary by module

Clause	Module	✅	🟡	❌	Standing
5(a)	System Design and Architecture	1	1	2	Partially compliant
5(b)(i)	Project Overview	3	0	0	**Compliant**
5(b)(ii)	Physical Progress Monitoring	3	0	1	Partially compliant
5(b)(iii)	Financial Monitoring	3	1	0	Substantially compliant
5(b)(iv)	Contract Management	1	3	0	Partially compliant
5(b)(v)	Risk and Alert System	0	2	1	**Weak**
5(b)(vi)	Reporting and Analytics	1	2	0	Partially compliant
5(b)(vii)	Data Security and Governance	2	1	1	Partially compliant
5(b)(viii)	Capacity Building and Support	0	1	2	**Weak**
4(c)–(e)	Consultant obligations needing system support	0	1	3	**Weak**

Prepared read-only against the BUIDCO dashboard codebase. No application file, schema object or configuration was altered in the course of this analysis.